20.3.4 Mutation Breeding

Mutation is defined as the sudden **heritable change** in the nucleotide sequence of DNA in an organism. It is a process by which **genetic variations** are created which in turn brings about changes in the organism. The organism which undergoes mutation is called a mutant.

The factors which induce mutations are known as mutagens or **mutagenic agents**. Mutagens are of two types namely physical mutagens and chemical mutagens.

More to Know

Gamma Garden

Gamma garden or Atomic garden is a concept popularised after World War II for the peaceful use of atomic energy for crop improvement. This is a type of induced mutation breeding where radioactive sources particularly gamma rays from Cobalt-60 or Caesium-137 are used to induce desirable mutations in crop plants.



i Physical mutagens

Radiations like X-rays, α, β and γ-rays, UV rays, temperature etc. which induce mutations are called physical mutagens

ii Chemical mutagens

Chemical substances that induce mutations are called chemical mutagens. e.g. Mustard gas and nitrous acid. The utilisation of induced mutation in crop improvement is called **mutation breeding**.

Achievements of mutation breeding

Some achievements of mutation breeding are

a. **Sharbati Sonora** wheat produced from Sonora-64 by using gamma rays.
b. **Atomita 2 rice** with saline tolerance and pest resistance
c. **Groundnuts** with thick shells

20.3.5 Hybridization

Hybridization may be defined as the process of crossing two or more types of plants for bringing their desired characters together into one progeny called **hybrid**. Hybrid is superior in one or more characters to both parents. Hybridization is the common method of creating genetic variation to get improved varieties.

Hybridization Experiment: *Triticale* (The first man – made cereal)

Triticale is the first man- made cereal hybrid. It is obtained by crossing wheat (*Triticum durum*, 2n = 28) and rye (*Secale cereal*, 2n = 14). The F_1 hybrid is sterile (2n = 21). Then the chromosome number is doubled using colchicine and it becomes a hexaploid *Triticale* (2n = 42).

The cycle of crop raising and selection continues till the plants with the desired characters are finally obtained. The development of new varieties is a long-drawn process. Two main aspects of hybridization are to combine the characters of two plants in one plant and to utilize hybrid vigour.

20.4 Animal Breeding

A **breed** is a group of animals of common origin within a species that has certain distinguishing characters that are not found in other members of the same species like general appearance and others striking features.

Breeding involves mating parents of different varieties each having some desired trait which are passed onto the offspring.

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